

[Подписка](#)[Спросить у чат-бота](#)[Игры и викторины](#)[История и общество](#)[Наука и технологии](#)[Биология](#)[Наука](#) > [Физика](#) > [Материя и энергия](#)[CITE](#)[Действия](#)

Определение местоположения и навигация

[↪](#) в [ультразвуковых](#)

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[Сонар](#) (звуковая навигация и определение дальности) широко применяется в морской сфере. Посылая импульсы [звука](#) или [ультразвука](#) и измеряя время, за которое они отражаются от удаленного объекта и возвращаются к источнику, можно [определить](#) местоположение этого объекта и отследить его движение. Этот метод широко используется для обнаружения и отслеживания [подводных лодок](#) в море, а также для поиска взрывных мин под поверхностью воды. Две [лодки](#), находящиеся в известных точках, также могут использовать метод триангуляции для определения местоположения и отслеживания третьей лодки или подводной лодки. Расстояние, на котором можно применять эти методы, ограничено температурными градиентами в воде, которые отклоняют луч от поверхности и создают теневые зоны. Одно из преимуществ ультразвуковых волн перед звуковыми в подводных системах заключается в том, что благодаря более высокой частоте (или более короткой длине волны) ультразвуковые волны распространяются на большие расстояния с меньшей дифракцией.



Эхолот также используется для составления карт дна океана, на основе которых составляются карты глубин, широко применяемые в навигации, особенно вблизи берегов и на мелководье. Даже небольшие лодки теперь оснащены эхолотами, которые определяют и отображают глубину воды, чтобы штурман мог избежать навала на подводные песчаные отмели или другие мелководные участки. Современные рыболовецкие суда используют ультразвуковые эхолоты для обнаружения косяков рыбы, что существенно повышает их эффективность.



Даже в отсутствие видимого света летучие мыши могут управлять своим полетом и даже находить летающих насекомых (которых они поедают в полете) с помощью звукового дальномера. Ультразвуковая эхолокация также используется в системах управления дорожным движением, а также для подсчета и сортировки предметов на

[конвейере](#). Ультразвуковой дальномер лежит в основе систем зрения роботов и имеет ряд важных медицинских применений (см. ниже).



Эффект Доплера

Если ультразвуковая [волна](#) отражается от движущегося препятствия, частота результирующей волны изменяется, то есть происходит доплеровский сдвиг. В частности, если препятствие движется по направлению к источнику, частота отраженной волны увеличивается, а если препятствие удаляется от источника, частота отраженной волны уменьшается. Величину частотного сдвига можно использовать для определения скорости движущегося препятствия. Точно так же, как доплеровский сдвиг в радиолокации, [электромагнитной волне](#), может быть использован для определения скорости движущегося автомобиля, скорость движущейся подводной лодки можно определить по доплеровскому сдвигу гидроакустического луча. Важным промышленным применением является ультразвуковой [расходомер](#), в котором отражение ультразвука от текущей жидкости приводит к доплеровскому сдвигу, который [калибруется](#) для определения скорости потока жидкости. Этот метод также применяется для измерения скорости кровотока в артериях.

Materials testing

Nondestructive testing involves the use of ultrasonic echolocation to gather information on the [integrity](#) of mechanical structures. Since changes in the material present an impedance mismatch from which an ultrasonic wave is reflected, ultrasonic testing can be used to identify faults, holes, cracks, or corrosion in materials, to inspect welds, to determine the quality of poured concrete, and to monitor [metal fatigue](#). Owing to the mechanism by which sound waves [propagate](#) in metals, ultrasound can be used to probe more deeply than any other form of radiation. Ultrasonic procedures are used to perform in-service inspection of structures in nuclear reactors.



Structural flaws in materials can also be studied by subjecting the materials to stress and looking for acoustic emissions as the materials are stressed. Acoustic emission, the general name for this type of nondestructive study, has developed as a distinct field of [acoustics](#).

High-intensity applications

High-intensity ultrasound has achieved a variety of important applications. Perhaps the most [ubiquitous](#) is ultrasonic [cleaning](#), in which ultrasonic vibrations are set up in small liquid tanks in which objects are placed for cleaning. Cavitation of the liquid by the ultrasound, as well as the vibration, create turbulence in the liquid and result in the cleaning action. Ultrasonic cleaning is very popular for jewelry and has also been used with such items as dentures, surgical instruments, and small machinery. Degreasing is often [enhanced](#) by ultrasonic cleaning. Large-scale ultrasonic cleaners have also been developed for use in assembly lines.



Ultrasonic [machining](#) employs the high-intensity vibrations of a transducer to move a [machine tool](#). If necessary, a slurry containing carborundum grit may be used; diamond tools can also be used. A variation of this technique is ultrasonic [drilling](#), which makes use of pneumatic vibrations at ultrasonic frequencies in place of the standard rotary drill bit. Holes of virtually any shape can be drilled in hard or brittle materials such as glass, germanium, or ceramic.

Ultrasonic [soldering](#) has become important, especially for soldering unusual or difficult materials and for very clean applications. The ultrasonic vibrations perform the function of cleaning the surface, even removing the oxide layer on aluminum so that the material can be soldered. Because the surfaces can be made extremely clean and free from the normal thin oxide layer, soldering flux becomes unnecessary.



Chemical and electrical uses

The chemical effects of ultrasound arise from an electrical discharge that accompanies the cavitation process. This forms a basis for ultrasound's acting as a [catalyst](#) in certain [chemical reactions](#), including oxidation, reduction, hydrolysis, polymerization and depolymerization, and molecular rearrangement. With ultrasound, some chemical processes can be carried out more rapidly, at lower temperatures, or more efficiently.



The ultrasonic delay line is a thin layer of piezoelectric material used to produce a short, precise delay in an electrical signal. The electrical signal creates a mechanical vibration in the piezoelectric crystal that passes through the crystal and is converted back to an electrical signal. A very precise time delay can be achieved by constructing a crystal with the proper thickness. These devices are employed in fast electronic timing [circuits](#).

Medical applications

Although [ultrasound](#) competes with other forms of [medical imaging](#), such as X-ray techniques and [magnetic resonance imaging](#), it has certain desirable features—for example, Doppler motion study—that the other techniques cannot provide. In addition, among the various modern techniques for the imaging of internal organs, ultrasonic devices are by far the least expensive. Ultrasound is also used for treating joint pains and for treating certain types of tumours for which it is desirable to produce localized heating. A very effective use of ultrasound deriving from its nature as a mechanical vibration is the elimination of kidney and bladder stones.



Diagnosis

Much medical [diagnostic](#) imaging is carried out with [X-rays](#). Because of the high photon energies of the X-ray, this type of radiation is highly ionizing—that is, X-rays are readily capable of destroying molecular bonds in the body [tissue](#) through which they pass. This destruction can lead to changes in the function of the tissue involved or, in extreme cases, its annihilation.

One of the important advantages of ultrasound is that it is a mechanical vibration and is therefore a nonionizing form of energy. Thus, it is usable in many sensitive circumstances where X-rays might be damaging. Also, the resolution of X-rays is limited owing to their great penetrating ability and the slight differences between soft tissues. Ultrasound, on the other hand, gives good contrast between various types of soft tissue.

Ultrasonic scanning in medical [diagnosis](#) uses the same principle as sonar. Pulses of high-frequency ultrasound, generally above one megahertz, are created by a piezoelectric transducer and directed into the body. As the ultrasound [traverses](#) various internal organs, it encounters changes in

[acoustic impedance](#), which cause [reflections](#). The amount and time delay of the various reflections can be analyzed to obtain information regarding the internal organs. In the B-scan mode, a linear array of transducers is used to scan a plane in the body, and the resultant data is displayed on a television screen as a two-dimensional plot. The A-scan technique uses a single transducer to scan along a line in the body, and the echoes are plotted as a function of time. This technique is used for measuring the distances or sizes of internal organs. The M-scan mode is used to record the motion of internal organs, as in the study of heart dysfunction. Greater resolution is obtained in ultrasonic imaging by using higher frequencies—i.e., shorter wavelengths. A limitation of this property of waves is that higher frequencies tend to be much more strongly absorbed.



Because it is nonionizing, ultrasound has become one of the staples of [obstetric](#) diagnosis. During the process of drawing amniotic fluid in testing for birth defects, ultrasonic imaging is used to guide the needle and thus avoid damage to the fetus or surrounding tissue. Ultrasonic imaging of the fetus can be used to determine the date of [conception](#), to identify multiple births, and to diagnose abnormalities in the development of the fetus.

Ultrasonic Doppler techniques have become very important in diagnosing problems in blood flow. In one technique, a three-megahertz ultrasonic beam is reflected off typical oncoming arterial blood with a Doppler shift of a few kilohertz—a frequency difference that can be heard directly by a physician. Using this technique, it is possible to monitor the heartbeat of a fetus long before a stethoscope can pick up the [sound](#). [Arterial diseases](#) such as arteriosclerosis can also be diagnosed, and the healing of arteries can be monitored following surgery. A combination of B-scan imaging and Doppler imaging, known as duplex scanning, can identify arteries and immediately measure their blood flow; this has been extensively used to diagnose heart valve defects.

Using ultrasound with frequencies up to 2,000 megahertz, which has a wavelength of 0.75 micrometre in soft tissues (as compared with a wavelength of about 0.55 micrometre for light), [ultrasonic microscopes](#) have

been developed that [rival](#) light microscopes in their resolution. The distinct advantage of ultrasonic microscopes lies in their ability to distinguish various parts of a cell by their viscosity. Also, because they require no artificial contrast mediums, which kill the cells, acoustic microscopy can study actual living cells.



Therapy and surgery

Because ultrasound is a mechanical vibration and can be well focused at high frequencies, it can be used to create internal heating of localized tissue without harmful effects on nearby tissue. This technique can be employed to relieve pains in joints, particularly in the back and shoulder. Also, research is now being carried out in the [treatment](#) of certain types of [cancer](#) by local heating, since focusing intense ultrasonic waves can heat the area of a tumour while not significantly affecting surrounding tissue.

Trackless [surgery](#)—that is, surgery that does not require an incision or track from the skin to the affected area—has been developed for several conditions. Focused ultrasound has been used for the treatment of [Parkinson's disease](#) by creating brain lesions in areas that are inaccessible to traditional surgery. A common application of this technique is the destruction of [kidney stones](#) with shock waves formed by bursts of focused ultrasound. In some cases, a device called an ultrasonic lithotripter focuses the ultrasound with the help of X-ray guidance, but a more common technique for destruction of kidney stones, known as endoscopic ultrasonic disintegration, uses a small metal rod inserted through the skin to deliver ultrasound in the 22- to 30-kilohertz frequency region.



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